COSC 4370 Homework #2

Overview:
Building on our work from the first assignment, we would like to be able to represent objects using triangles and to show more complex transformations. To demonstrate these ideas, we will be creating a simplified representation of the inner solar system. This will make use of different translation and rotation matrices as well as the ability to push and pop these transformations. We would like the movements to be somewhat representative of the real inner solar system but scale the sizes to be a pleasing representation.

Specifications:
Starting with the code from Homework #1, generate a simplified representation of the inner solar system. Begin by constructing a yellow sphere of **radius 2.0** at the origin [0,0,0] to represent the sun. Position the Earth, represented by a **radius 1.0** blue sphere at **an orbital radius of 10** from the origin in the *xy*-plane. Add spheres for Mercury (red), Venus (green), and Mars (red) in the *xy*-plane. These spheres should be scaled based on their relative size to the Earth and position at the correct relative distance from the Sun. (Note that the planet scale and orbital scale are different to avoid a substantial amount of empty space). Position a Moon at a **radius of 1.5** around the Earth and scaled proportionally to the Earth.

Set the visualization in motion, with the Earth moving at some base rate (an angle of one degree per update is smooth). Set the orbital periods for the other planets relative to the speed for the Earth. Add in controls so that up-arrow and down-arrow keypresses will rotate the projection plane (see the video below). The limits of rotation should be from directly above the orbital plane down to the orbital plane (90 degrees).

Deliverables:
- Submit a single python file to Canvas, including the required specifications.
- The program should display a colored sphere for sun, each of the inner planets, and the moon.
- Provide a mechanism to rotate the visual projection from directly above the orbital plane to the orbital plane using up and down arrow keypresses.

Notes:
- Be sure to enable the depth test, or your spheres will not correctly occlude each other.
- In pygame, the up and down arrow keypresses are:
  pygame.K_UP and pygame.K_DOWN
- Use the code you developed for the first Homework to generate this simple demonstration of the orbits of the inner planets with the relative scales given below. The scales for the size of the spheres will be different than the scale for the orbits of the spheres.

	Period, days	Radius, km	Relative Size	Orbital radius (x 10^6 km)	AU
Mercury	**87.97**	**2440**	**0.38**	**57.9**	**0.39**
Venus	**224.70**	**6052**	**0.95**	**108.2**	**0.72**
Earth	**365.26**	**6371**	**1.00**	**149.6**	**1**
Moon	**27.3**	**1738**	**0.27**	**0.384****	

Mars	**686.98**	**3390**	**0.53**	**228**	**1.5**

** Use a radius of 1.5 for the orbit of the moon.